

Std - X

Sub - Science



CHAPTER 1
**Chemical Reactions
and Equations**





CHAPTER 1

Chemical Reactions and Equations

Part – 3

Types of Chemical Reactions

Types of Chemical Reactions

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graph TD; A[Types of Chemical Reactions] --- B[1) Combination Reaction]; A --- C[2) Decomposition Reaction]; A --- D[3) Displacement Reaction]; A --- E[4) Double Displacement Reaction]; A --- F[5) Oxidation and Reduction (Redox) Reaction];
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1) Combination Reaction

2) Decomposition Reaction

3) Displacement Reaction

4) Double Displacement Reaction

5) Oxidation and Reduction (Redox) Reaction

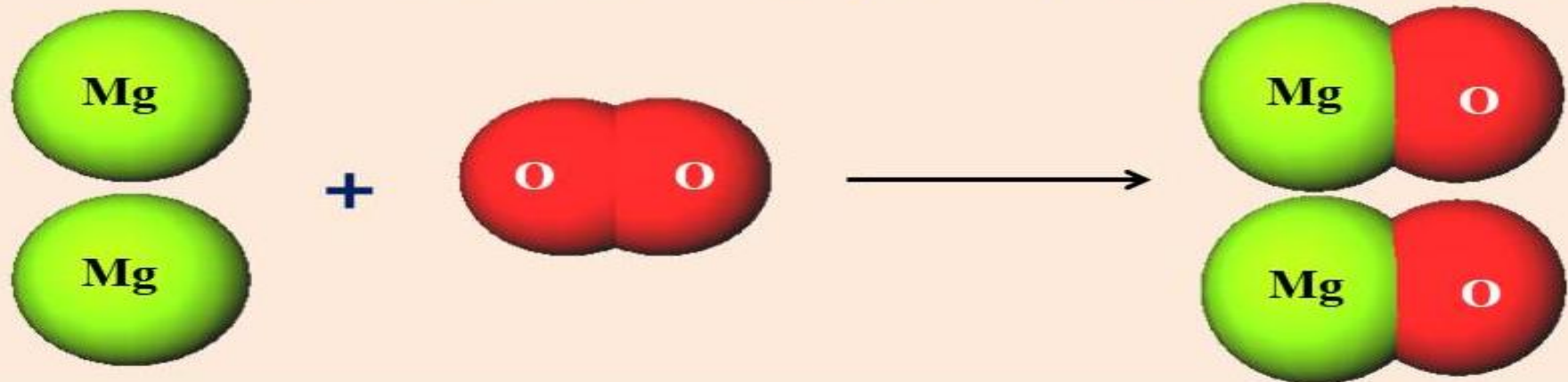
Types of Chemical Reactions



Chemical reactions involve the breaking and making of bonds between atoms to produce new substances.



Atomic model of balanced chemical equation (for illustration purpose)



A) Combination Reaction

A chemical reaction in which **two or more** reactants combine to produce a **'single'** product is known as **'combination reaction'**.



A) Combination Reaction

Activity 1.4 : Reaction between Calcium Oxide and Water



CaO(s)
(Quicklime)



+ $\text{H}_2\text{O(l)}$
(Water)

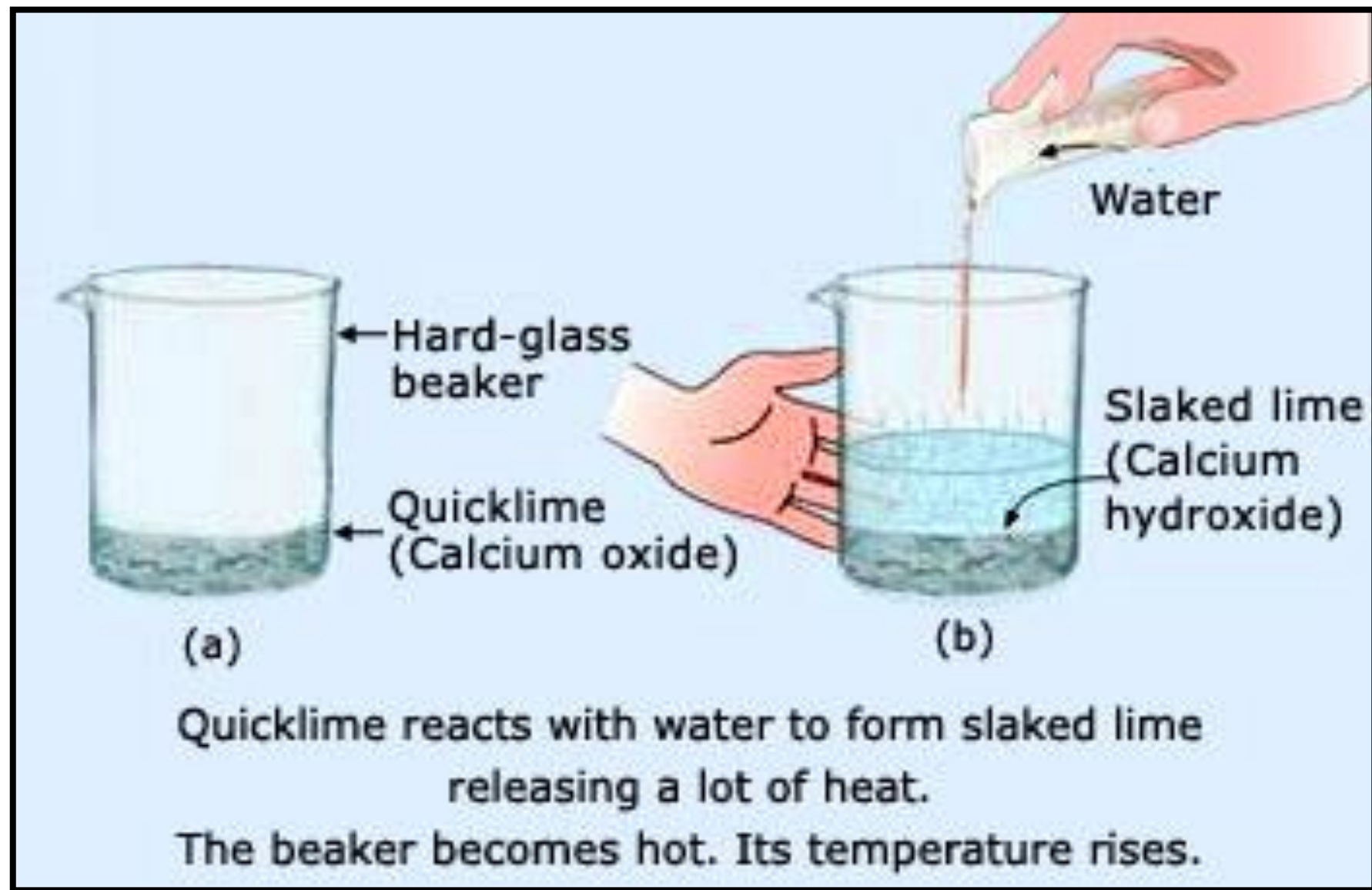


$\text{Ca(OH)}_2\text{(aq)}$
(Slaked lime)

In this reaction, **Calcium oxide** and **water** combine to form a single product '**calcium hydroxide**'. So this is a combination reaction.

A) Combination Reaction

Be Careful !
This is
**'Exothermic
Reaction'** as a
large amount
of heat is
produced and
beaker
becomes hot.



Did You Know?

- ✓ A solution of slaked lime is used for whitewashing walls.
- ✓ Calcium hydroxide reacts slowly with the carbon dioxide in air to form a thin layer of calcium carbonate on the walls.
- ✓ Calcium carbonate is formed after two to three days of whitewashing and gives a shiny finish to the walls.
- ✓ It is interesting to note that the chemical formula for marble is also CaCO_3 .



Calcium
hydroxide

Carbon
Dioxide

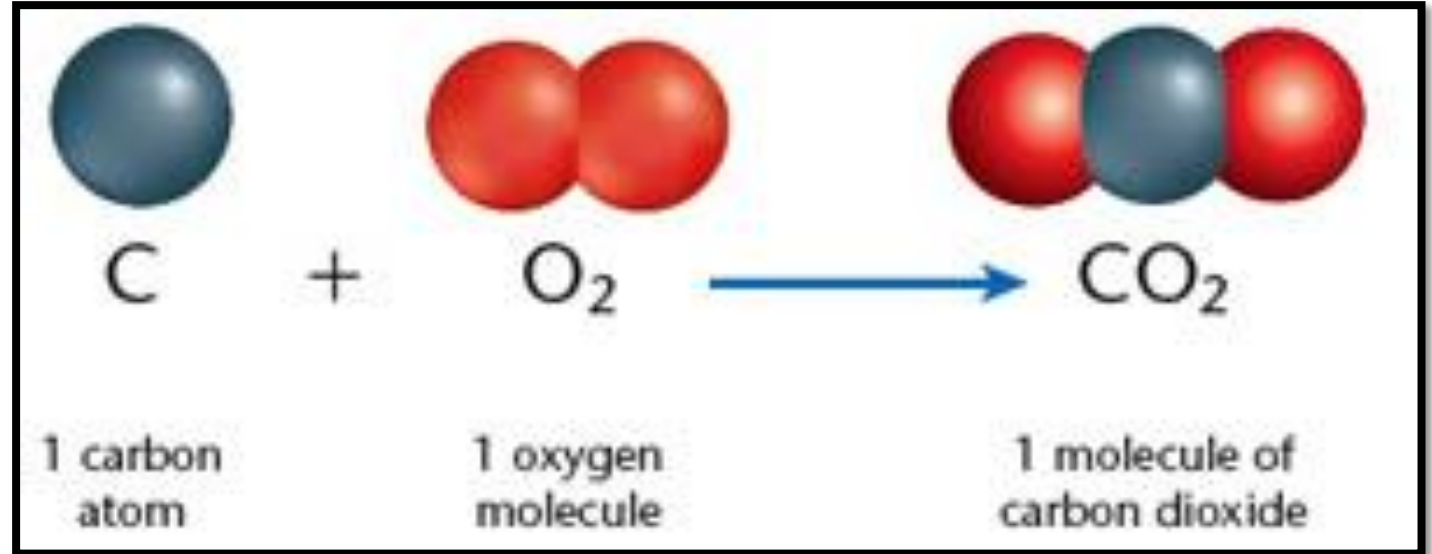
Calcium
carbonate

Water



Some More Examples of combination reactions

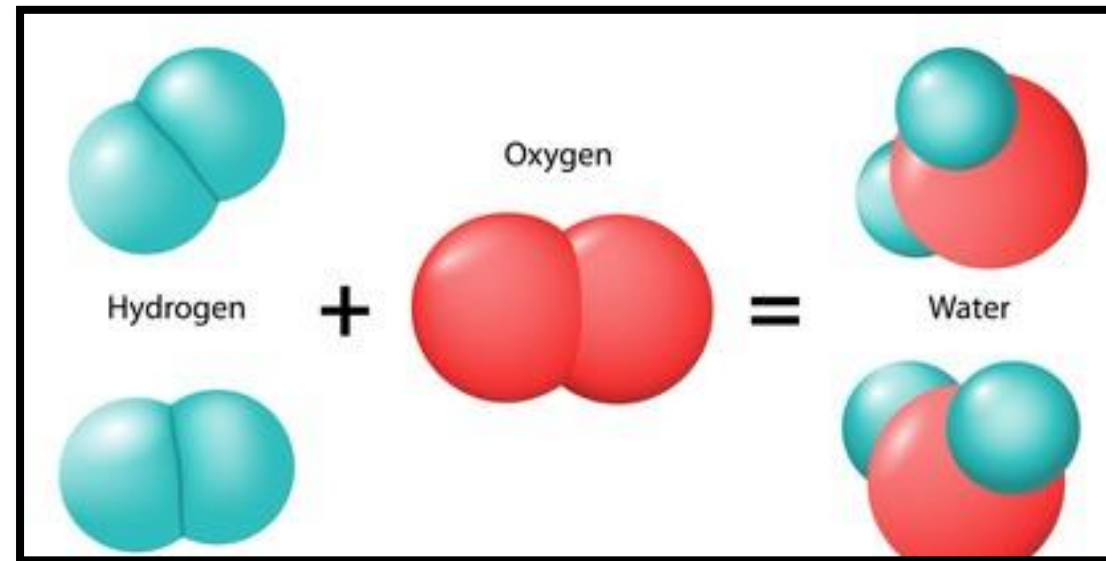
1) Burning of Coal



- A large amount of heat is evolved while coal is burning.
- Reactions in which heat is released along with the formation of products are called exothermic chemical reactions.

Some More Examples of combination reactions

2) Formation of water from 'Hydrogen' gas and 'oxygen' gas

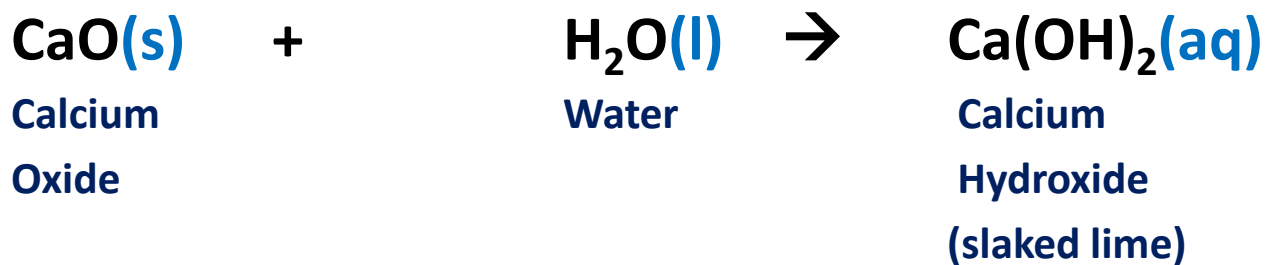


Exothermic Reaction

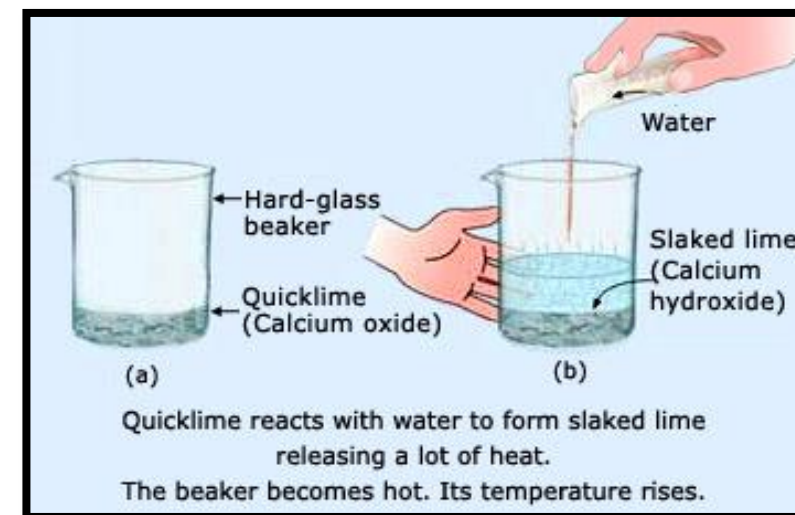
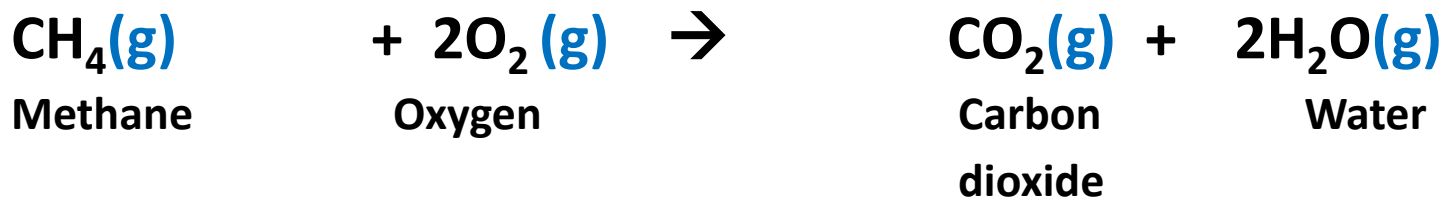
A chemical reaction in which heat is released along with the formation of products is known as 'exothermic reaction'.

Examples of exothermic reaction :

1) Formation of slaked lime



2) Burning of natural gas –



3) Respiration is an Exothermic Reaction

- We need energy to stay alive.
- We get this energy from the food we eat.
- During digestion, food is broken down into simpler substances. E.g. rice, potatoes and bread contain carbohydrates.
- These carbohydrates are broken down to form glucose.
- This glucose combines with oxygen in the cells of our body and provides energy (heat).
- The name of this chemical reaction is 'respiration'.

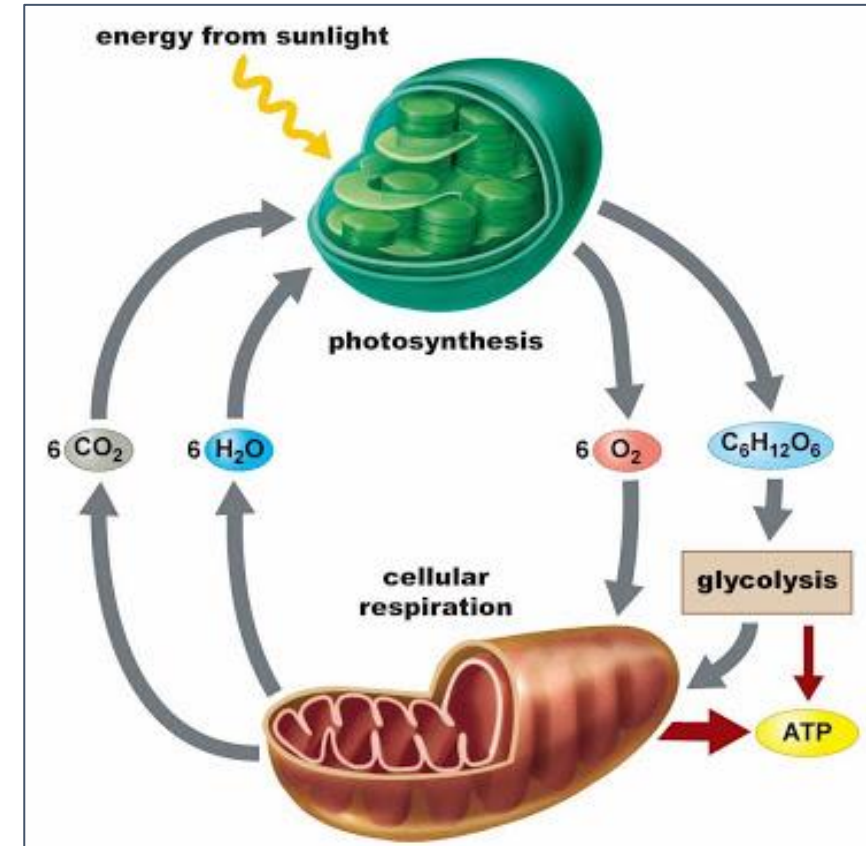


Glucose

Oxygen

Carbon
dioxide

Water



Cellular Respiration

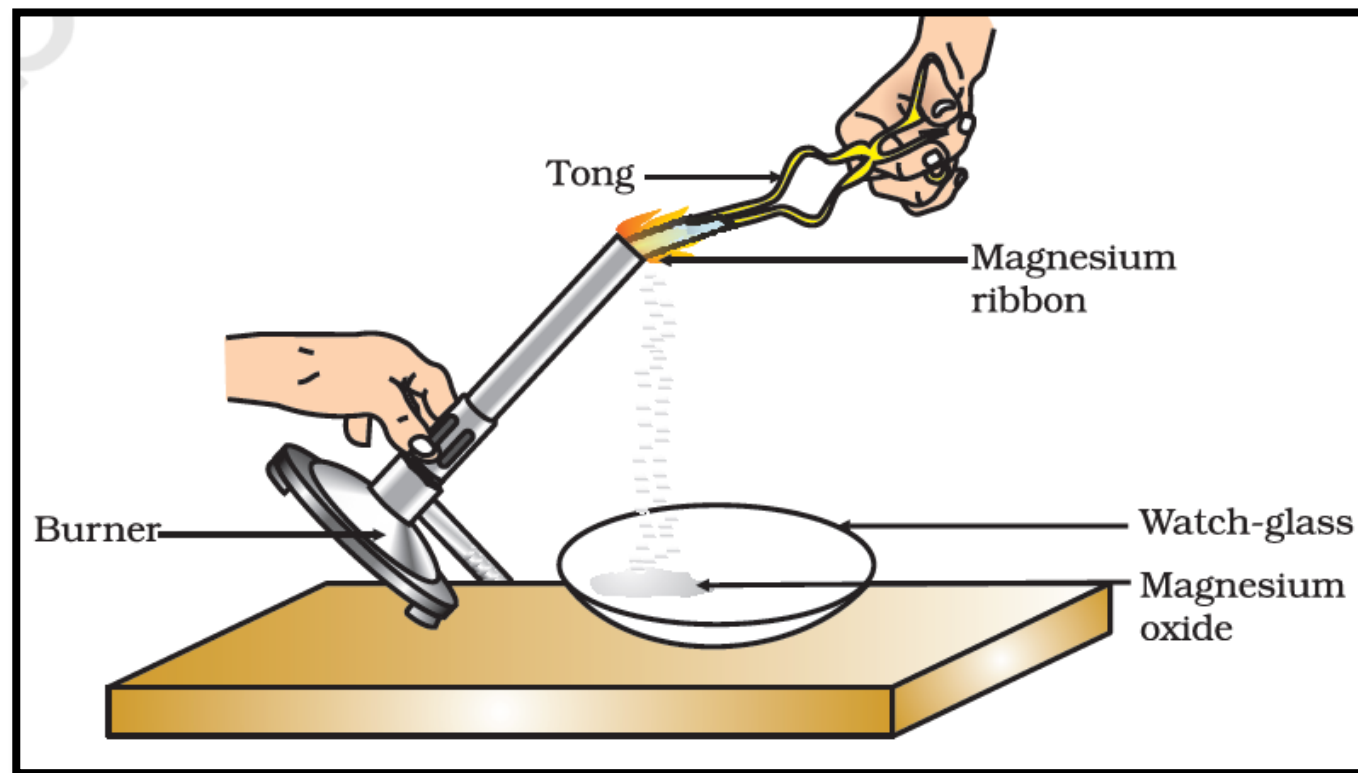
Exothermic Reaction

4) The decomposition of vegetable matter into compost is also an example of an 'exothermic reaction'.



Identify the type of reaction in the following.

- **Burning of Magnesium Ribbon in air.**



- Two reactants combine to form a single product. So it is 'Combination reaction'.
- During burning of magnesium ribbon, a lot of heat is evolved. So it is an 'exothermic reaction'.

TO BE CONTINUED

